Binary classification for 'distributed system' related terminology, using Word to Vector and Back Propagation Models.

Summary

To create this model, Word to Vector (W2V) was used to produce a dataset for binary classification, using Back Propagation. Words classified as either "related" or "not related" to the concept of a "distributed system" and was trained on text from the book "DISTRIBUTED SYSTEMS" by Maarten van Steen Andrew S. Tanenbaum.

This text was chosen because the language was likely to be strongly associated with this concept. The expectation was that a strong output of words most similar to "distributed" and "system" would be produced by the W2V model, serving as a good foundational dataset for Back Propagation training.

Word to Vector Model

Model Training: The model was trained using Gensim W2V, assigning 300 vectors to each word and only learning words appearing more than 20 times. 1360 words were identified, of which the top sixty were extracted to train the Back Propagation model. (fig.1)

```
Model trained!
The word 'system' is represented by 300 numbers:
[ 0.09670141  0.18278165  0.18709768  0.12304901  0.01794064 -0.07113019
  0.19484061  0.11267492 -0.14360413  0.21159019] ... (showing first 10 of 300)
```

```
[7]: #Data is formated into a table.
vocab_all = model.wv.index_to_key
vectors_all = [model.wv[i] for i in vocab_all]

dsys_all_data = pd.DataFrame(vectors_all)
dsys_all_data.index = vocab_all
print(f"We have {len(vocab_all)} words, each with {len(vectors_all[0])} numbers")
print(f"Example words: {vocab_all[:10]}")

# Find words with similar embeddings (similar contexts in DS4)
# These are words that appear in similar situations to "system" and "distributed"

print("Words most similar to 'system' based on context in Distributed Systems 4:")
model.wv.most_similar(['system'], topn=60) #Finds top 60 words most similar to subject

# Create DataFrame with custom column names
sys_top60 = pd.DataFrame(model.wv.most_similar(['system'], topn=60), columns=['word', 'similarity'])
```

```
We have 1340 words, each with 300 numbers
Example words: ['system', 'server', 'process', 'message', 'node', 'client', 'data', 'may', 'example', 'distributed']
Words most similar to 'system' based on context in Distributed Systems 4:
```

```
[8]: dsys_all_data.head()
```

[8]:

	0	1	2	3	4	5	6	7	8	9	...	290	291	292	
system	0.096701	0.182782	0.187098	0.123049	0.017941	-0.071130	0.194841	0.112675	-0.143604	0.211590	...	-0.023611	0.213789	0.117151	0.
server	0.037712	0.090536	0.137370	0.032959	0.057070	-0.306244	0.172445	0.362906	0.234070	-0.042550	...	0.038796	0.139714	0.300315	-0.
process	0.037319	0.067845	0.111354	0.101779	0.002439	-0.259925	0.243635	0.364641	0.188855	-0.015957	...	-0.017585	0.146154	0.307226	-0.
message	0.016393	0.017219	0.099021	0.085131	0.034786	-0.309837	0.168164	0.392657	0.252865	-0.044263	...	0.024978	0.099767	0.305491	-0.
node	0.021870	0.098012	0.167264	0.085789	0.069818	-0.255062	0.183401	0.355938	0.189060	-0.007223	...	0.012100	0.154441	0.228849	0.

5 rows × 300 columns

Figure 1: Word to vector model results.

Back Propagation Model

Model Training: The Back Propagation model was trained using the medium SpaCy NLP and a neural network which had 5 neurons within the hidden layer. The model took a list of words, “related” and “not related”. The “related” words were initially generated using W2V based on the input words “distributed” and “system”, then assigned a class of “1” for “related” and “0” for “not related”.

Model Predictions:

```
words = [

# Good words 60
'distributed', 'computing', 'computer', 'many', 'security', 'page', 'important', 'often', 'network',
'architecture', 'design', 'service', 'operating', 'solution', 'application', 'mechanism', 'protocol',
'approach', 'problem', 'conference', 'survey', 'issue', 'general', 'algorithm', 'implementation', 'th',
'component', 'international', 'peer-to-peer', 'discussed', 'model', 'mobile', 'decentralized',
'symposium', 'information', 'networked', 'web', 'performance', 'ieee', 'much', 'internet', 'software',
'organization', 'generally', 'way', 'acm', 'difficult', 'virtual', 'various', 'discus', 'replication',
'described', 'provided', 'make', 'fault', 'open', 'scheme', 'caching', 'principle', 'technique',

# Bad words 75
'item', 'object', 'thing', 'element', 'entity', 'unit', 'factor', 'aspect', 'part', 'section', 'area',
'field', 'category', 'type', 'form', 'kind', 'level', 'stage', 'step', 'process', 'method', 'result',
'outcome', 'value', 'number', 'amount', 'size', 'length', 'width', 'height', 'volume', 'weight', 'rate',
'ratio', 'range', 'limit', 'boundary', 'point', 'position', 'location', 'place', 'space', 'time', 'period',
'interval', 'moment', 'date', 'year', 'month', 'day', 'instance', 'example', 'case', 'sample', 'set',
'group', 'list', 'collection', 'series', 'line', 'order', 'sequence', 'pattern', 'structure', 'pot',
'model', 'rule', 'standard', 'measure', 'index', 'record', 'entry', 'detail', 'table', 'sculpture'

]
```

Dataset 1

For the first iteration of training, I used a set of words that was extracted from the W2V model and a set of neutral words randomly generated by an LLM. (fig. 2)

Figure 2: Training dataset 1.

```
Total params: 1,511 (5.90 KB)

Trainable params: 1,511 (5.90 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 665us/step - accuracy: 0.5238 - loss: 0.7009
Epoch 2/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 675us/step - accuracy: 0.5619 - loss: 0.6427
Epoch 3/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 685us/step - accuracy: 0.7429 - loss: 0.6103
Epoch 4/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 668us/step - accuracy: 0.8095 - loss: 0.5853
Epoch 5/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 638us/step - accuracy: 0.8190 - loss: 0.5590
Epoch 6/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 610us/step - accuracy: 0.8095 - loss: 0.5293
Epoch 7/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 584us/step - accuracy: 0.8476 - loss: 0.5002
Epoch 8/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 599us/step - accuracy: 0.8667 - loss: 0.4736
Epoch 9/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 592us/step - accuracy: 0.8667 - loss: 0.4463
Epoch 10/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 609us/step - accuracy: 0.8952 - loss: 0.4232
Epoch 11/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 592us/step - accuracy: 0.8857 - loss: 0.4026
Epoch 12/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 628us/step - accuracy: 0.8952 - loss: 0.3846
Epoch 13/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 599us/step - accuracy: 0.8952 - loss: 0.3689
Epoch 14/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 680us/step - accuracy: 0.9048 - loss: 0.3541
Epoch 15/15
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 570us/step - accuracy: 0.9048 - loss: 0.3400
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step

Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.67      0.74        24
           1       0.69      0.86      0.77        21

    accuracy                           0.76        45
   macro avg       0.77      0.76      0.76        45
weighted avg       0.77      0.76      0.75        45
```

The training model used 15 epochs initially with a loss of 0.3400.

```
[18]: pred('pond')
      1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step
[18]: {'related': 0.783702552318573}
```

Figure 3: Training dataset 1 results.

(fig.3) When inputting the word ‘pond’ into the model, an incorrect result was returned. This indicated that the word list would need to be revised. Words such as 'th', 'los', 'way' were removed and words like 'scalability', 'api', 'authentication', 'synchronization', 'cluster' were added.

words = [

 # Good words 75
'distributed', 'computing', 'computer', 'many', 'security', 'page', 'important', 'often', 'network', 'architecture', 'design', 'service', 'operating', 'solution', 'application', 'mechanism', 'protocol', 'approach', 'problem', 'conference', 'survey', 'issue', 'general', 'algorithm', 'implementation', 'component', 'international', 'peer-to-peer', 'discussed', 'model', 'mobile', 'decentralized', 'symposium', 'information', 'networked', 'web', 'performance', 'ieee', 'much', 'internet', 'software', 'organization', 'generally', 'acm', 'difficult', 'virtual', 'various', 'discus', 'replication', 'described', 'provided', 'make', 'fault', 'open', 'scheme', 'caching', 'principle', 'technique', 'system', 'way', 'communication', 'developed', 'cloud', 'lecture', 'scalability', 'api', 'authentication', 'synchronization', 'cluster', 'encryption', 'decryption', 'middleware', 'interface', 'framework', 'database'

 # Bad words 75
'apple', 'river', 'mountain', 'forest', 'ocean', 'desert', 'flower', 'butterfly', 'sunshine', 'lake', 'cloud', 'thunder', 'breeze', 'island', 'valley', 'meadow', 'tree', 'leaf', 'stone', 'sand', 'shell', 'coral', 'fish', 'bird', 'feather', 'nest', 'garden', 'harvest', 'farmer', 'village', 'bridge', 'castle', 'paint', 'sculpture', 'music', 'dance', 'poetry', 'novel', 'theatre', 'camera', 'photograph', 'canvas', 'palette', 'guitar', 'piano', 'violin', 'drum', 'flute', 'ballet', 'costume', 'festival', 'candle', 'lantern', 'blanket', 'pillow', 'chair', 'table', 'window', 'curtain', 'carpet', 'sea', 'wood', 'glass', 'shirt', 'cup', 'pineapple', 'kangaroo', 'lavender', 'waterfall', 'cinnamon', 'butterfly', 'glacier', 'peacock', 'sunflower', 'rainbow'

]

Figure 4: Training Data set 2.

Dataset 2:

(fig.4) For the second data set the number of "Good words" was increased to 75 and new set of "Bad words were used, that were polar opposite to the "good words.

```
Total params: 1,511 (5.90 KB)

Trainable params: 1,511 (5.90 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 648us/step - accuracy: 0.5524 - loss: 0.7037
Epoch 2/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 705us/step - accuracy: 0.7619 - loss: 0.5545
Epoch 3/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 693us/step - accuracy: 0.8762 - loss: 0.4659
Epoch 4/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 664us/step - accuracy: 0.8952 - loss: 0.3968
Epoch 5/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 585us/step - accuracy: 0.9238 - loss: 0.3350
Epoch 6/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 598us/step - accuracy: 0.9429 - loss: 0.2892
Epoch 7/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 571us/step - accuracy: 0.9524 - loss: 0.2493
Epoch 8/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 596us/step - accuracy: 0.9619 - loss: 0.2126
Epoch 9/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 606us/step - accuracy: 0.9810 - loss: 0.1815
Epoch 10/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 621us/step - accuracy: 0.9810 - loss: 0.1571
Epoch 11/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 602us/step - accuracy: 0.9905 - loss: 0.1388
Epoch 12/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 593us/step - accuracy: 0.9905 - loss: 0.1245
Epoch 13/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 638us/step - accuracy: 1.0000 - loss: 0.1127
Epoch 14/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 616us/step - accuracy: 1.0000 - loss: 0.1034
Epoch 15/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 639us/step - accuracy: 1.0000 - loss: 0.0955
Epoch 16/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 637us/step - accuracy: 1.0000 - loss: 0.0888
Epoch 17/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 597us/step - accuracy: 1.0000 - loss: 0.0836
Epoch 18/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 588us/step - accuracy: 1.0000 - loss: 0.0782
Epoch 19/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 575us/step - accuracy: 1.0000 - loss: 0.0739
Epoch 20/20
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 580us/step - accuracy: 1.0000 - loss: 0.0705
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step
```

This time the training model used 20 epochs and produced a loss of 0.0705.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.57      0.70        23
           1       0.68      0.95      0.79        22

    accuracy                           0.76        45
   macro avg       0.80      0.76      0.75        45
weighted avg       0.81      0.76      0.75        45

[24]: pred('pond')

      1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step
[24]: {'related': 0.6280168890953064}
```

(fig.5) The result was still incorrect and the score hadn't changed that much meaning that the problem was likely with the list of "Bad words", mostly within one narrow domain specific category of nature.

Figure 5: Training dataset 2 results.

```
words = [

    # consolidated list of 75 words that are related to "distributed" and "system" (Good words) (1)
    'distributed', 'computing', 'computer', 'many', 'security', 'screen', 'important', 'often', 'network',
    'architecture', 'design', 'service', 'operating', 'solution', 'application', 'mechanism', 'protocol',
    'approach', 'problem', 'conference', 'survey', 'issue', 'algorithm', 'implementation', 'component',
    'international', 'peer-to-peer', 'discussed', 'mobile', 'decentralized', 'symposium', 'networked',
    'web', 'performance', 'ieee', 'much', 'internet', 'software', 'organization', 'acm', 'difficult',
    'virtual', 'discus', 'replication', 'described', 'provided', 'make', 'fault', 'open', 'scheme',
    'caching', 'principle', 'technique', 'system', 'hub', 'communication', 'developed', 'cloud',
    'node', 'scalability', 'api', 'authentication', 'synchronization', 'cluster', 'encryption',
    'decryption', 'middleware', 'interface', 'framework', 'database', 'consensus', 'throughput',
    'partition', 'quorum', 'sharding',

    # consolidated list of 130 words that have no relation to "distributed" and "system" (Bad words) (0)
    'apple', 'banana', 'orange', 'strawberry', 'pineapple', 'carrot', 'potato', 'onion', 'tomato',
    'dog', 'cat', 'horse', 'cow', 'bird', 'eagle', 'penguin', 'tree', 'flower', 'grass', 'river',
    'ocean', 'beach', 'mountain', 'sun', 'moon', 'cloud', 'rain', 'snow', 'pizza', 'burger', 'sushi',
    'pasta', 'coffee', 'tea', 'milk', 'cake', 'chocolate', 'icecream', 'shirt', 'pants', 'dress',
    'shoe', 'hat', 'desk', 'chair', 'sofa', 'bed', 'lamp', 'mirror', 'clock', 'fork', 'spoon',
    'plate', 'bowl', 'canvas', 'brush', 'paint', 'pottery', 'clay', 'sculpture', 'guitar', 'piano',
    'drum', 'song', 'concert', 'heart', 'lung', 'smile', 'laugh', 'cry', 'love', 'friend', 'family',
    'baby', 'soccer', 'tennis', 'dance', 'yoga', 'movie', 'actor', 'theater', 'car', 'bus', 'train',
    'bicycle', 'airplane', 'ticket', 'hotel', 'road', 'bridge', 'red', 'blue', 'green', 'yellow',
    'circle', 'square', 'triangle', 'big', 'small', 'tall', 'short', 'pillow', 'blanket', 'towel',
    'soap', 'shampoo', 'toothbrush', 'lipstick', 'comb', 'wallet', 'keys', 'umbrella', 'backpack',
    'suitcase', 'passport', 'screwdriver', 'camera', 'photograph', 'candle', 'birthday',
    'book', 'newspaper', 'pencil', 'pen', 'eraser', 'page', 'staple', 'sword', 'gun', 'lance'
]
```

Figure 6: Training dataset 3.

Dataset 3:

(fig.6) For the third iteration of training, 75 “related” and 130 “not related” words were used to train the model. Some words were removed and new domain specific “Good words” were added to the list such as ‘hub’ and ‘node’. The bad words were revised with samples of multiple domain specific words added.

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 5)	1,505
dense_1 (Dense)	(None, 1)	6

Total params: 1,511 (5.90 KB)

Trainable params: 1,511 (5.90 KB)

Non-trainable params: 0 (0.00 B)

```
Epoch 1/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 626us/step - accuracy: 0.3217 - loss: 0.7688
Epoch 2/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 658us/step - accuracy: 0.5175 - loss: 0.7033
Epoch 3/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 628us/step - accuracy: 0.6294 - loss: 0.6536
Epoch 4/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 523us/step - accuracy: 0.6923 - loss: 0.6000
Epoch 5/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 518us/step - accuracy: 0.8392 - loss: 0.5327
Epoch 6/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 501us/step - accuracy: 0.8951 - loss: 0.4554
Epoch 7/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 505us/step - accuracy: 0.9161 - loss: 0.3878
Epoch 8/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 490us/step - accuracy: 0.9091 - loss: 0.3340
Epoch 9/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 491us/step - accuracy: 0.9231 - loss: 0.2898
Epoch 10/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 512us/step - accuracy: 0.9301 - loss: 0.2537
Epoch 11/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 499us/step - accuracy: 0.9301 - loss: 0.2252
Epoch 12/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 508us/step - accuracy: 0.9371 - loss: 0.2008
Epoch 13/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 556us/step - accuracy: 0.9650 - loss: 0.1788
Epoch 14/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 545us/step - accuracy: 0.9650 - loss: 0.1618
Epoch 15/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 558us/step - accuracy: 0.9510 - loss: 0.1492
Epoch 16/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 566us/step - accuracy: 0.9650 - loss: 0.1332
Epoch 17/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 528us/step - accuracy: 0.9650 - loss: 0.1218
Epoch 18/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 566us/step - accuracy: 0.9720 - loss: 0.1130
Epoch 19/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 563us/step - accuracy: 0.9650 - loss: 0.1053
Epoch 20/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 531us/step - accuracy: 0.9720 - loss: 0.0964
Epoch 21/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 550us/step - accuracy: 0.9650 - loss: 0.0907
Epoch 22/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 534us/step - accuracy: 0.9720 - loss: 0.0865
Epoch 23/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 553us/step - accuracy: 0.9720 - loss: 0.0818
Epoch 24/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 569us/step - accuracy: 0.9720 - loss: 0.0764
Epoch 25/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 522us/step - accuracy: 0.9720 - loss: 0.0745
Epoch 26/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 526us/step - accuracy: 0.9720 - loss: 0.0733
Epoch 27/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 566us/step - accuracy: 0.9650 - loss: 0.0677
Epoch 28/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 519us/step - accuracy: 0.9650 - loss: 0.0667
Epoch 29/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 527us/step - accuracy: 0.9720 - loss: 0.0637
Epoch 30/30
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 562us/step - accuracy: 0.9580 - loss: 0.0614
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.95      0.92        40
           1       0.89      0.77      0.83        22

    accuracy                           0.89        62
   macro avg       0.89      0.86      0.87        62
weighted avg       0.89      0.89      0.89        62

Layer: dense
Weights:
[[-0.3527667  -0.53625125 -0.05205517 -0.23245567  0.08767648]
 [-0.04111394 -0.10027532  0.07879476  0.17750065 -0.09921747]
 [-0.07053572 -0.18572378 -0.05522016 -0.07823381 -0.08503815]
 ...
 [-0.02611319  0.01921873 -0.14876139 -0.07168691  0.17578706]
 [ 0.11999536  0.06988294  0.10656844  0.02606266  0.01373909]
 [ 0.23105943  0.1963585   0.09522544  0.10726137 -0.00140317]]
Biases:
[0.23681694 0.40241715 0.04375712 0.06518871 0.0116585 ]

Layer: dense_1
Weights:
[[-0.6338027 ]
 [-0.5173192 ]
 [ 1.0782878 ]
 [ 0.3724218 ]
 [ 0.90485984]]
Biases:
[-0.1473874]
```

This time the training model used 30 epochs and produced a loss of 0.0614.

```
[14]: prediction('pond')

      1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step

[14]: {'not related': 0.23620645701885223}
```

Figure 7: Training dataset 3 results.

(fig. 7) The result was correct and the score had reduced by almost 0.4 points.

To test the data further, a greater volume of data will need to be used.

Testing on Unseen Data

To further test the model, I built a bulk processing feature to make predictions on large numbers of words at the same time. I extracted the most used and highest similarity scoring words from each of the documents which were taken from the UK government website for technology transfer (listed in the document column of fig. 8 and 9). Listed in fig. 8 are the top 50 related results by number of times used and fig. 9 are listed by similarity score.

This data set could be further developed combining the count across publications to find the most common related language and filtered to words with a similarity score above 0.9. This would likely produce a data set of common language with high relatedness; the danger however could mean that the model becomes over fit.

1	word	count	prediction	similarity	document
2	business	416	related	0.989	Uk industrial strategy policy paper 2026
3	network	391	related	0.9874	Realising the benefits of 5G
4	consumer	345	related	0.8435	Realising the benefits of 5G
5	use	339	related	0.9439	Realising the benefits of 5G
6	impact	337	related	0.992	Realising the benefits of 5G
7	web	331	related	0.9666	Web3 App Comprehensive Research Book
8	case	312	related	0.5237	Realising the benefits of 5G
9	investment	303	related	0.989	Uk industrial strategy policy paper 2026
10	group	288	related	0.9482	Realising the benefits of 5G
11	sector	278	related	0.989	Uk industrial strategy policy paper 2026
12	growth	249	related	0.9315	Uk industrial strategy policy paper 2026
13	new	244	related	0.8435	Uk industrial strategy policy paper 2026
14	government	241	related	0.8502	Uk industrial strategy policy paper 2026
15	industry	233	related	0.7012	Uk industrial strategy policy paper 2026
16	strategy	229	related	0.9702	Uk industrial strategy policy paper 2026
17	industrial	227	related	0.9912	Uk industrial strategy policy paper 2026
18	scenario	214	related	0.9769	Realising the benefits of 5G
19	innovation	182	related	0.9894	Uk industrial strategy policy paper 2026
20	econometrics	180	related	0.9976	Realising the benefits of 5G
21	across	174	related	0.9264	Uk industrial strategy policy paper 2026
22	technology	170	related	0.9912	Uk industrial strategy policy paper 2026
23	study	147	related	0.9924	Realising the benefits of 5G
24	energy	142	related	0.9189	Uk industrial strategy policy paper 2026
25	skill	141	related	0.9739	Uk industrial strategy policy paper 2026
26	transaction	140	related	0.8161	Web3 App Comprehensive Research Book
27	application	134	related	0.987	Realising the benefits of 5G
28	national	134	related	0.7554	Uk industrial strategy policy paper 2026
29	opportunity	133	related	0.5944	Uk industrial strategy policy paper 2026
30	mobile	131	related	0.8004	Realising the benefits of 5G
31	deployment	124	related	0.9946	Realising the benefits of 5G
32	service	123	related	0.9848	Uk industrial strategy policy paper 2026
33	economy	122	related	0.8246	Uk industrial strategy policy paper 2026
34	model	119	related	0.9993	Realising the benefits of 5G
35	economic	119	related	0.8246	Uk industrial strategy policy paper 2026
36	contract	117	related	0.685	Web3 App Comprehensive Research Book
37	asset	113	related	0.989	Web3 App Comprehensive Research Book
38	global	110	related	0.9729	Uk industrial strategy policy paper 2026
39	market	108	related	0.9436	Uk industrial strategy policy paper 2026
40	token	107	related	0.6365	Web3 App Comprehensive Research Book
41	result	106	related	0.9376	Realising the benefits of 5G
42	billion	106	related	0.5814	Uk industrial strategy policy paper 2026
43	user	105	related	0.9003	Realising the benefits of 5G
44	million	105	related	0.5814	Uk industrial strategy policy paper 2026
45	data	104	related	0.9976	Uk industrial strategy policy paper 2026
46	plan	102	related	0.6906	Uk industrial strategy policy paper 2026
47	might	100	related	0.8982	Realising the benefits of 5G
48	also	99	related	0.5524	Uk industrial strategy policy paper 2026
49	time	98	related	0.8982	Realising the benefits of 5G
50	year	98	related	0.5418	Uk industrial strategy policy paper 2026

Figure 8: Top 50 results by No. times used.

1	word	count	prediction	similarity	document
2	model	119	related	0.9993	Realising the benefits of 5G
3	based	83	related	0.9993	Realising the benefits of 5G
4	approach	64	related	0.9993	Uk industrial strategy policy paper 2026
5	reform	57	related	0.9993	Uk industrial strategy policy paper 2026
6	transparency	36	related	0.9993	Web3 App Comprehensive Research Book
7	implementation	36	related	0.9993	Realising the benefits of 5G
8	governance	34	related	0.9993	Web3 App Comprehensive Research Book
9	protocol	34	related	0.9993	Web3 App Comprehensive Research Book
10	stakeholder	28	related	0.9993	Realising the benefits of 5G
11	empirical	19	related	0.9993	Realising the benefits of 5G
12	concept	17	related	0.9993	Web3 App Comprehensive Research Book
13	foundational	16	related	0.9993	Web3 App Comprehensive Research Book
14	paradigm	14	related	0.9993	Web3 App Comprehensive Research Book
15	econometrics	180	related	0.9976	Realising the benefits of 5G
16	data	104	related	0.9976	Uk industrial strategy policy paper 2026
17	critical	59	related	0.9967	Uk industrial strategy policy paper 2026
18	environmental	41	related	0.9967	Realising the benefits of 5G
19	assessment	38	related	0.9967	Realising the benefits of 5G
20	participant	15	related	0.9967	Web3 App Comprehensive Research Book
21	education	15	related	0.9967	Web3 App Comprehensive Research Book
22	audit	14	related	0.9967	Web3 App Comprehensive Research Book
23	primary	13	related	0.9967	Web3 App Comprehensive Research Book
24	consensus	43	related	0.9966	Web3 App Comprehensive Research Book
25	malicious	12	related	0.9965	Web3 App Comprehensive Research Book
26	priority	53	related	0.9963	Uk industrial strategy policy paper 2026
27	fundamentally	17	related	0.9963	Web3 App Comprehensive Research Book
28	browser	19	related	0.996	Web3 App Comprehensive Research Book
29	continuous	15	related	0.9957	Web3 App Comprehensive Research Book
30	engine	13	related	0.9955	Web3 App Comprehensive Research Book
31	deployment	124	related	0.9946	Realising the benefits of 5G
32	logistics	40	related	0.9946	Realising the benefits of 5G
33	intervention	44	related	0.9941	Uk industrial strategy policy paper 2026
34	assumption	70	related	0.9933	Realising the benefits of 5G
35	purpose	26	related	0.9933	Realising the benefits of 5G
36	principle	23	related	0.9933	Web3 App Comprehensive Research Book
37	authority	19	related	0.9933	Web3 App Comprehensive Research Book
38	study	147	related	0.9924	Realising the benefits of 5G
39	research	62	related	0.9924	Uk industrial strategy policy paper 2026
40	speed	45	related	0.9924	Uk industrial strategy policy paper 2026
41	analysis	44	related	0.9924	Realising the benefits of 5G
42	capacity	36	related	0.9924	Realising the benefits of 5G
43	power	33	related	0.9924	Web3 App Comprehensive Research Book
44	improvement	30	related	0.9924	Realising the benefits of 5G
45	efficiency	26	related	0.9924	Realising the benefits of 5G
46	partnership	91	related	0.9921	Uk industrial strategy policy paper 2026
47	autonomous	13	related	0.9921	Web3 App Comprehensive Research Book
48	impact	337	related	0.992	Realising the benefits of 5G
49	verifiable	26	related	0.9917	Web3 App Comprehensive Research Book
50	relevant	21	related	0.9917	Realising the benefits of 5G

Figure 9: Top 50 results by similarity score.